FROM RESEARCH TO INDUSTRY

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UML INTRODUCTION

Introduction on UML for Industrial Systems

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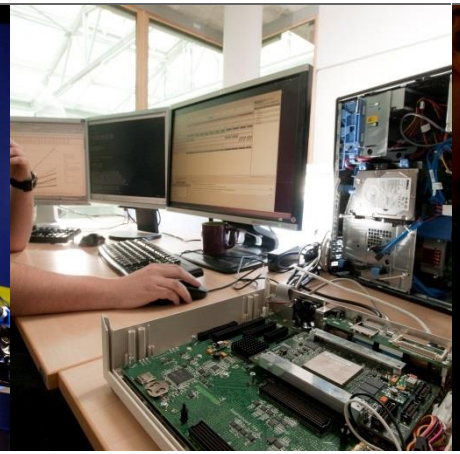

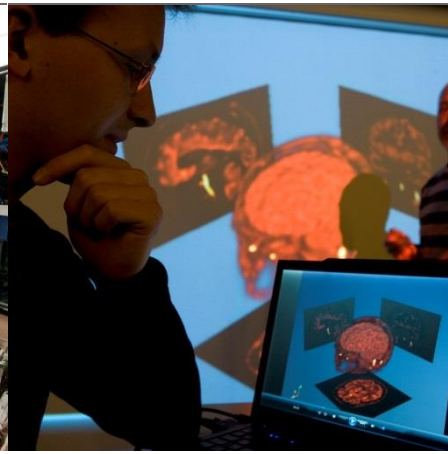

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OUTLINE

UML design

- Abstract syntax
- Notation
- Semantics

UML

- What is UML ?
- What is the purpose of UML

CONTENT

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UNIFIED MODELING LANGUAGE

A. Language = syntax + semantics

- Syntax
  - Rules by which language elements (e.g. words) are assembled into expressions (e.g., phrases, clauses)
- Semantics
  - Rules by which syntactic expressions are assigned to their meanings

B. Generic and great expressiveness

C. Abstract syntax and notation

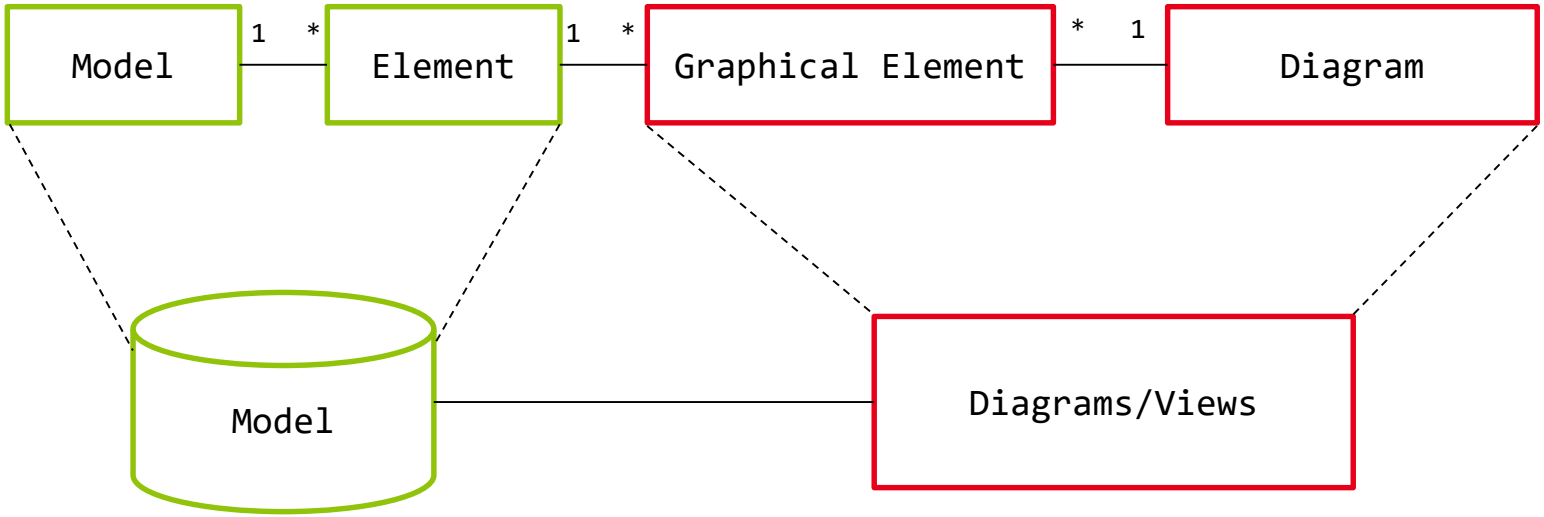
- The set of language concepts is the abstract syntax
- Elements in this set may or may not have a representation
  - The representation is what is shown to the designer which manipulates the language concepts through a tool.

Rules for naming, scoping, visibility, integrity + execution (limited)

Examples

- Semantic rule: Class [1]
  - English: If a Class is concrete, all the Operations of the Class should have a realizing Method.
  - OCL:
```
not self.isAbstract implies self.allOperations->
                                forAll (op | self.allMethods->
                                             exists (m | m.specification-> includes(op)))
```
- Syntactic rule: Class
  - Basic notation : “A class is drawn as a solid-outline rectangle with three compartments separated by horizontal lines.”
  - Presentation option: “Either or both of the attribute and operation compartments may be suppressed.”

ABSTRACT SYNTAX DEFINITION



A. Interpretation

- When you delete an element from the model you delete all its graphical elements from the diagram.
- When you delete a graphical element, you DO NOT necessarily delete the corresponding element in the model.

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UML SCOPE

BENEFITS AND LIMITATIONS

A unique formalism for any application type

- Database design
- Embedded systems design
- Multimedia
- Information systems, etc.

But UML stays at the language level

- It does not provide any development process / methodology

  - Development tasks organization ?
  - Responsibility distribution ?
  - Basics to use in an appropriate manner ?

UML is:

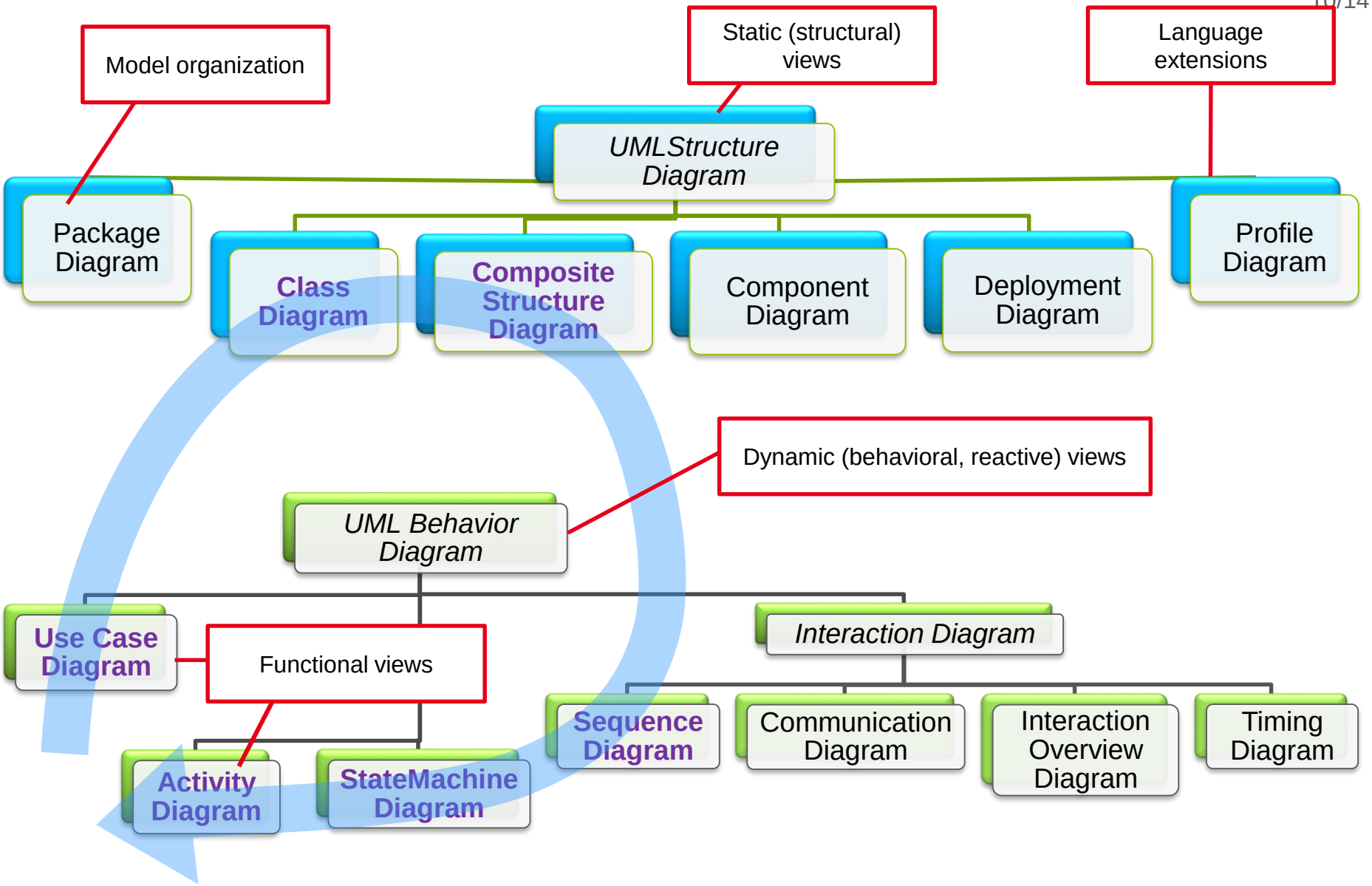
- a notation
- adequate for all of the object methods
- a modeling language
- free and public

UML is the standard notation to build object models, architectures and to describe behaviors

UML is not

- a methodology
- necessarily object oriented

UML DIAGRAMS



A UML BASED APPROACH

Synthesis of several viewpoints

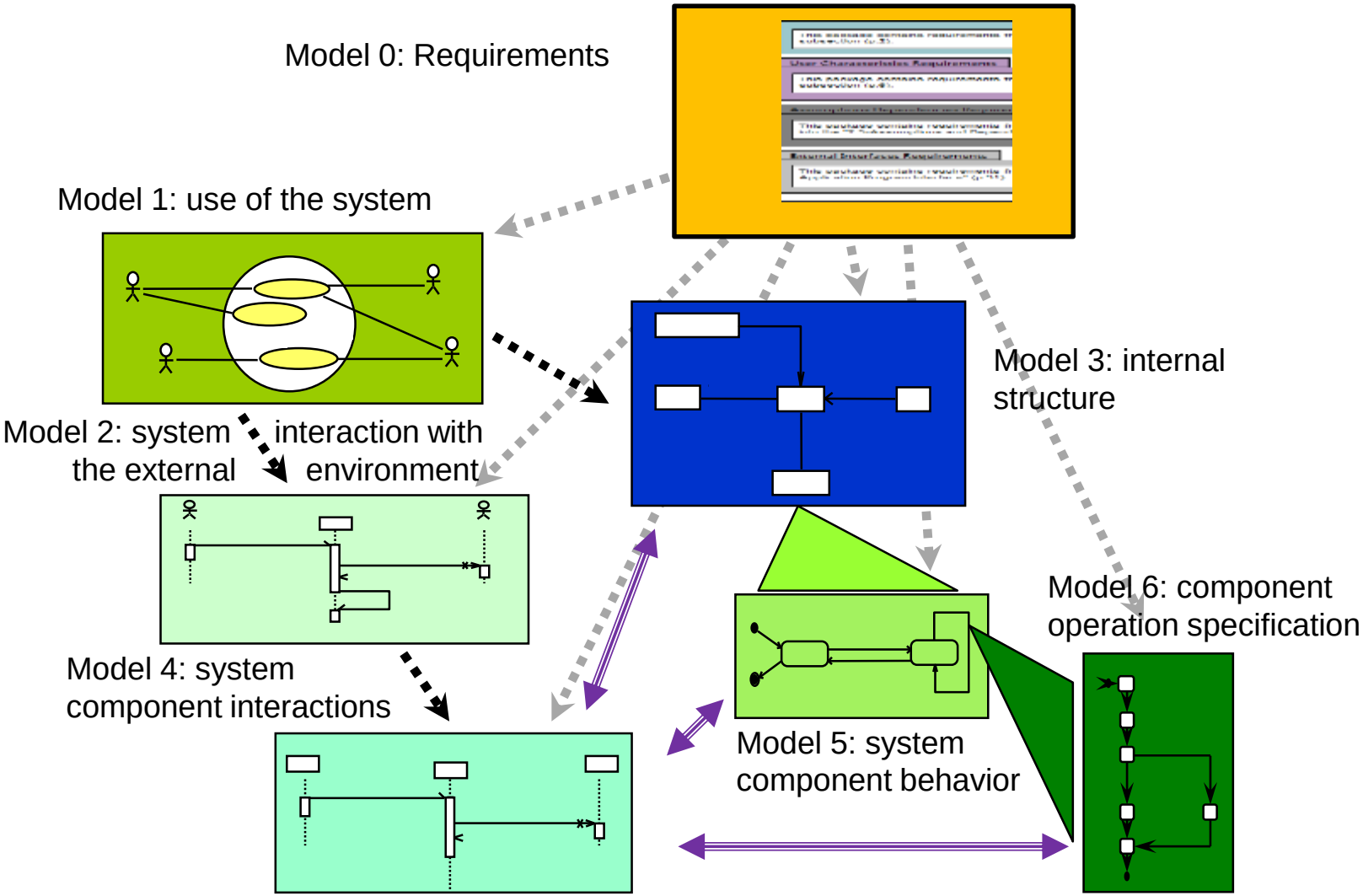
- Functional: usage and algorithms
- Static: structure
- Dynamic: reactive behaviors and interactions

Ensure consistency between viewpoints

Incremental approach of the development process from problem analysis to system implementation through iterative refinements of the system model (or set of models) using the same formalism.

- No language discontinuity
- Possibility of continuity among the tools

COMPLEMENTARY MODELS FOR COMPLEMENTARY VIEWS OF A COMPLETE MODEL IN AN INCREMENTAL APPROACH



- **Consistency rules to ensure non ambiguous modeling**
- **Formal analysis of the models becomes possible**

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